

```
open(FILE, "<\"", "$fh") or die "Cannot open $fh: $!";
open(MAIL, "|/usr/sbin/sendmail -t");
print MAIL "FROM: $from\n";
print MAIL "TO: $to\n";
print MAIL "Subject: $subject\n";
print MAIL "Content-Type: text/html; charset=us-ascii\n";
print MAIL "<FILE>";
close(FILE);
close(MAIL);
```

Emailing via PHP is well explained in the following URL for all the above three cases, though it does not include style sheets: http://webcheatsheet.com/php/send_email_text_html_attachment.php



Note: There are two major differences between PHP and Shell/Perl when it comes to effective emailing:

- PHP is itself embedded HTML, while in Perl we have to separately embed to make it HTML.
- PHP has a built-in mail function, which does a major chunk of the mailing. We could very well write a Perl module or Shell function file that we could reuse or call in programs. I prefer a Perl module because of the level of intricacies that can be included to make it a perfect function; and then we can use it within a Shell program too.

Continued from page 26...

```
remnux@remnux:~$ cat bptrans
Level 2:
+ xor_add: XOR with 8 bits static key A, then ADD with 8 bits static key B. Parameters: A (1-FF), B (1-FF).
+ add_xor: ADD with 8 bits static key A, then XOR with 8 bits static key B. Parameters: A (1-FF), B (1-FF).
+ xor_inc: XOR with 8 bits key A incrementing after each character. Parameters: A (0-FF).
+ xor_dec: XOR with 8 bits key A decrementing after each character. Parameters: A (0-FF).
+ sub_inc: SUB with 8 bits key A incrementing after each character. Parameters: A (0-FF).
+ xor_chained: XOR with 8 bits key A chained with previous character. Parameters: A (1-FF).
+ xor_rotated: XOR with 8 bits key A chained with next character (Reverse order from end to start). Parameters: A (1-FF).
Level 3:
+ xor_inc_rot: XOR with 8 bits key A incrementing after each character.
```

Figure 6: Bptrans

```
remnux@remnux:~$ xorsearch -s malware.exe smtp
Found XOR 00 position 350160: smtp1ib(.
Found XOR 00 position 3FC140: smtp.p....#?....Y...f....2.J...f...
....>.HTP.
Found XOR F1 position 91715: smtp.....j...v.....[9...x[.L.jo...
addB.B.p.
Found ADD 02 position 370E11: smtp.....BQ.Ie.....9Hq.....Iv..1.V.U
...0..U....
```

Figure 7: Xorsearch

bptrans.py -l 2 malware.exe

Xorsearch

Xorsearch tools help to find all possible 1 byte key values. However, the particular string that we are looking for is



Figure 7: Mail with HTML contrary to normal text



References

- <https://en.wikipedia.org/wiki/MIME>
- http://www.perlmonks.org/?node_id=675595 à Net::SMTP
- <http://perldoc.perl.org/MIME/Base64.html>
- <http://search.cpan.org/~mivkovic/Mail-Sendmail-0.79/Sendmail.pm>
- <http://search.cpan.org/~rjbs/MIME-Lite-3.030/lib/MIME/Lite.pm>
- <http://www.if-not-true-then-false.com/2012/vi-vim-syntax-highlighting-on-fedora-centos-red-hat-rhel/>
- <http://www.vim.org/download.php>
- <http://perl.plover.com/local.html>

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known and one good value is 'smtp', because some malware samples often send mail to the command and control server.

The syntax is:

Xorsearch [option] [filename.extension] <string>

An example is:

Xorsearch -s malware.exe smtp

References

- <https://en.wikipedia.org/wiki/Malware>
- www.threattracksecurity.com/enterprise-security/malware-analysis-sandbox-software.aspx
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- <http://www.decacalge.info/python/balbuzard>

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